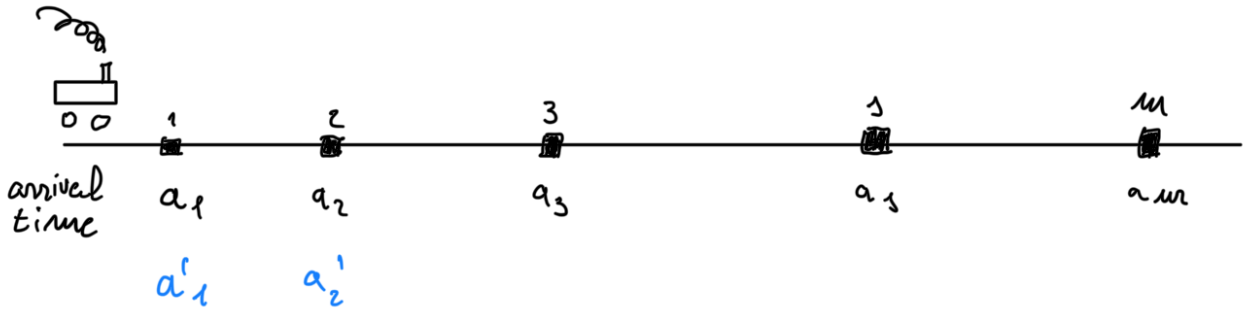
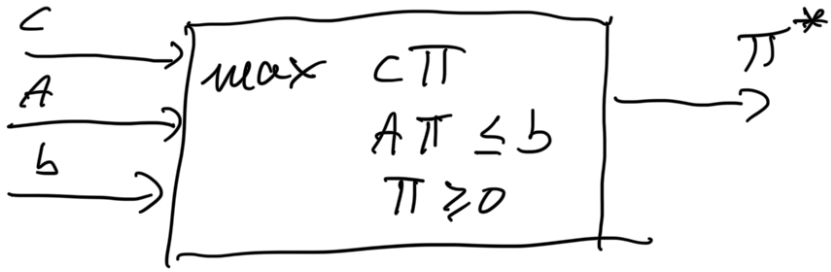


$a'_1 = a_1 + \Pi_1$ $\qquad$ $a'_s = a_s + \Pi_s$

$$\Pi_s = \begin{cases} < 0 & \text{early} \\ = 0 & \text{no damage} \\ > 0 & \text{delay} \end{cases}$$



$$\Pi_s = \underset{\geq 0}{\Pi_s^+} - \underset{\geq 0}{\Pi_s^-}$$

Investment problem

x_i = amount of bond i

$$x_1 + x_2 + x_3 + x_4 + x_5 \leq 137$$

$x_1 \leq 40$ a_1 $\qquad$ $x_4 \leq 5$ a_4

$x_2 \leq 12$ a_2 $\qquad$ $x_5 \leq 400$ a_5 $\qquad$ $x_i \geq 0$, $i = 1..5$

$x_3 \leq 130$ a_3

$$\max \quad 3.2\frac{x_1}{40} + 1.5\frac{x_2}{12} + 4.2\frac{x_3}{130} + 0.7\frac{x_4}{5} + 17\frac{x_5}{400}$$

$0 \le y_i \le 1$ relative value of bond i w.r.t the availability

$$y_i = \frac{x_i}{a_i}$$

$$\max \quad 3.2\, y_1 + 1.5\, y_2 + 4.2\, y_3 + 0.7\, y_4 + 17\, y_5$$

$$40 y_1 + 12 y_2 + 130 y_3 + 5 y_4 + 400 y_5 \le 137$$

$$0 \le y_i \le 1 \quad i = 1..5$$

$$\frac{3.2}{40} \quad \frac{1.5}{12} \quad \frac{4.2}{130} \quad \frac{0.7}{5} \quad \frac{17}{400}$$

3 2 5 1 4

bond	a_i		budget	y_i	return
4	5	$\leftrightarrow$	137	1	0.7
2	12	$\leftrightarrow$	132	1	1.5
1	40	$\leftrightarrow$	120	1	3.2
5	400	$\leftrightarrow$	80	$\frac{80}{400} = \frac{1}{5}$	$17 \cdot \frac{1}{5} = 3.4$
			0		
3				0	

burglar problem

$$x_i = \begin{cases} 1 & \text{item } i \text{ taken} \\ 0 & \text{item } i \text{ discarded} \end{cases}$$

$$\max \sum_{i=1}^{n} v_i \, x_i$$

$$\sum_{i=1}^{n} w_i \, x_i \le b$$

7 2 4 5 4 1

5	3	2	3	1	1
4	6	2	3	1	5

item	w_i	b	x_i	v_i
5	1	8	1	4
3	2	7	1	4
4	3	5	1	5
1	5	2	0	0
6	1	2	1	1
2	~~3~~	1	0	0
				$\boxed{14}$

$x_1 = 1 \quad x_3 = 1 \quad x_5 = 1$

$$x \in \{ \underset{y_1}{v_1} \; \cdots \; \underset{y_r}{v_r} \}$$

$$x = \sum_{i=1}^{r} v_i y_i$$

$$\sum_{i=1}^{r} y_i = 1$$

$$v_1 < v_2 < \quad < v_r$$

$$x = v_1 y_1 + (v_2 - v_1) y_2 + \quad + (v_r - v_{r-1}) y_r$$

$y = [1 \; 1 \; 1 \; 0 \; 0 \; 0 \; 0]$ $\qquad x = v_3$

$[1 \; 0 \; 1 \; 0 \; 1 \; 0 \; 0 \; 1]$ $\qquad$ NO

$x = ?$

y_i	$\leq$	y_{i-1}
0		0
0,1		1